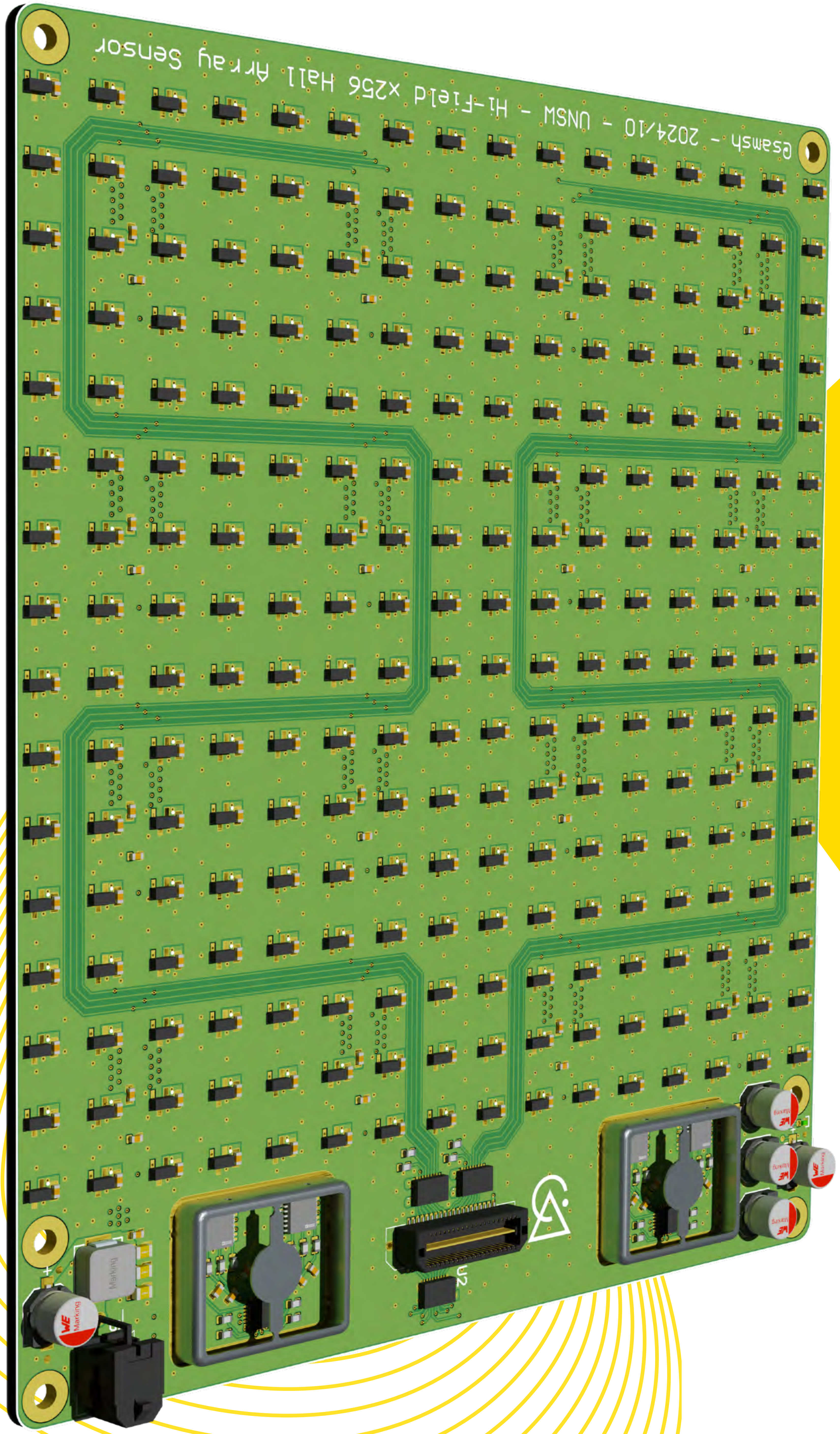


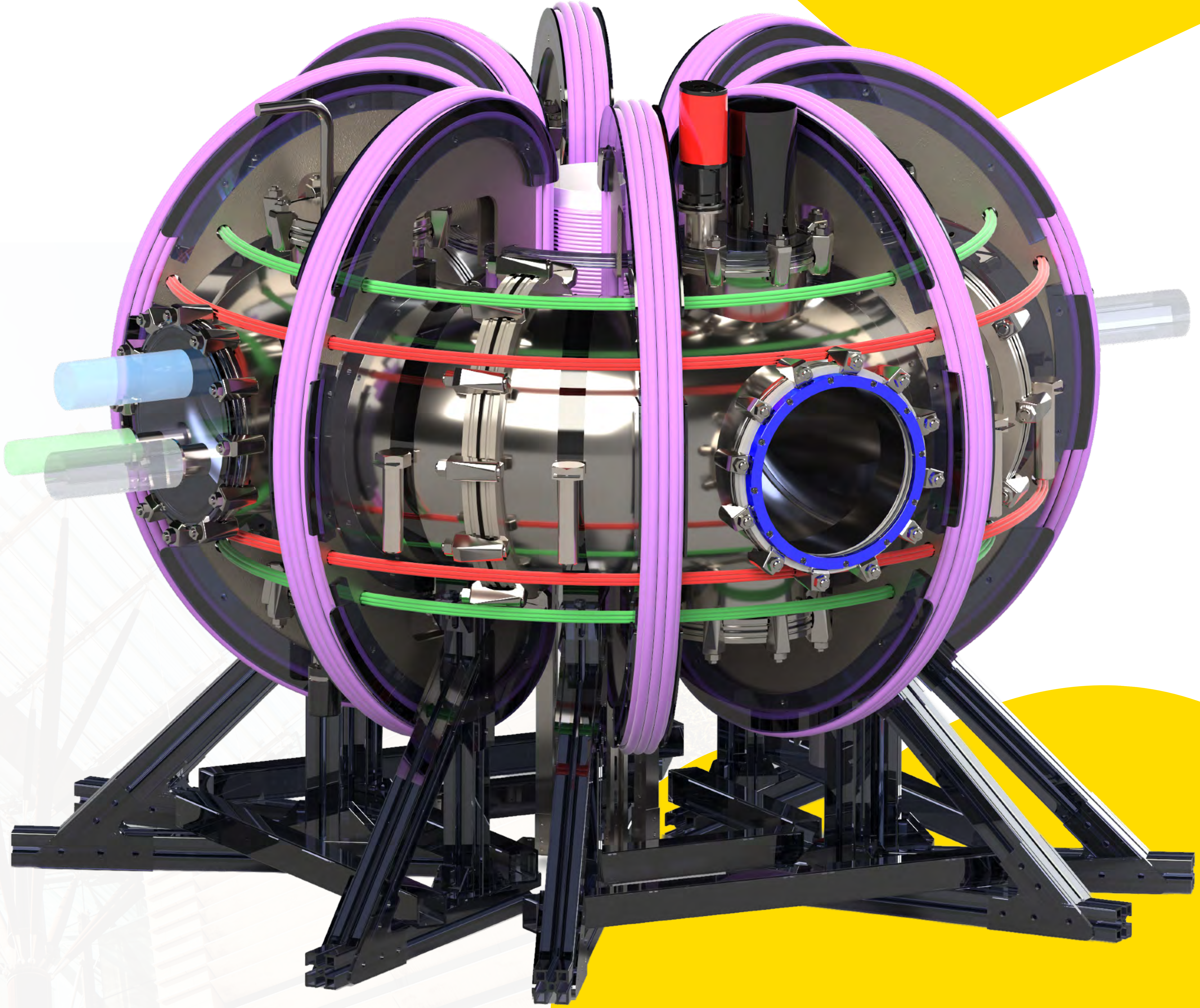
Hall Arrays for 2D Tokamak Magnetic Field Imaging

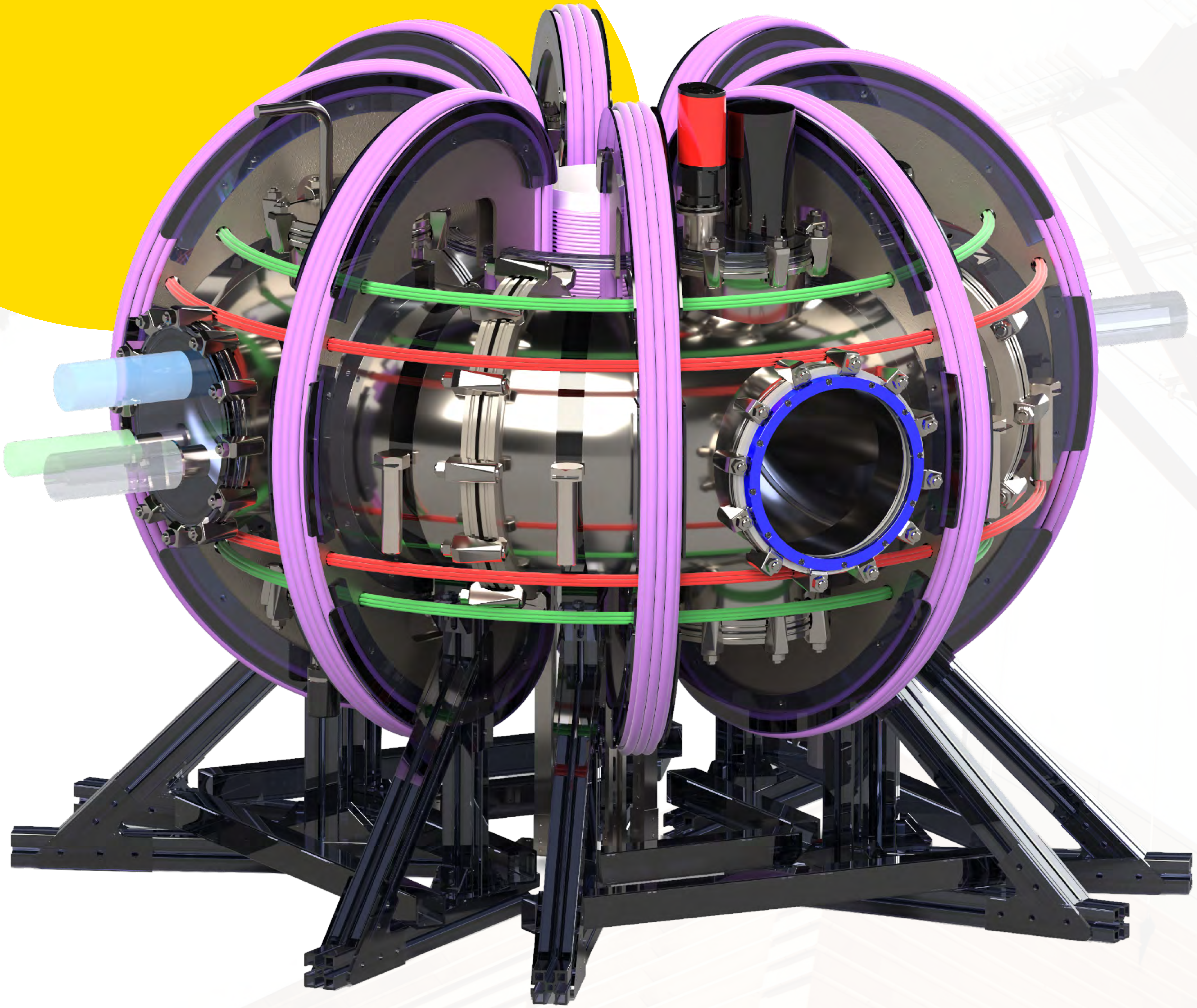
Sam Shelton, Thesis C Conference, August 2025

UNSW SYDNEY

The SOUTH Tokamak

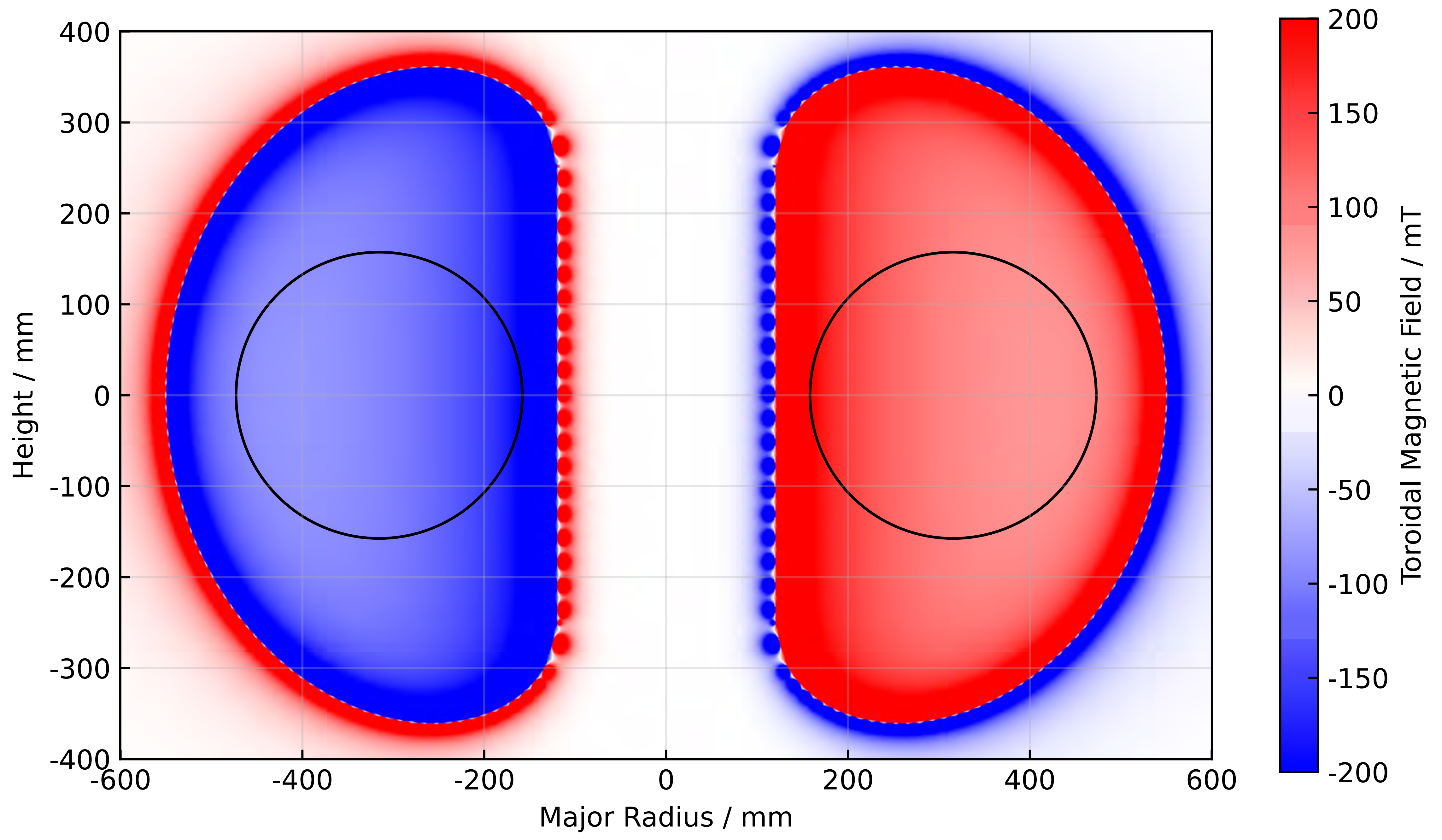
Constructed as part of the ATOMCRAFT vertically integrated project, the Student Operated Undergraduate Tokamak for Hydrogen (SOUTH) will be the worlds first fusion device of its class designed and built by undergraduates when it is switch on in late 2026.



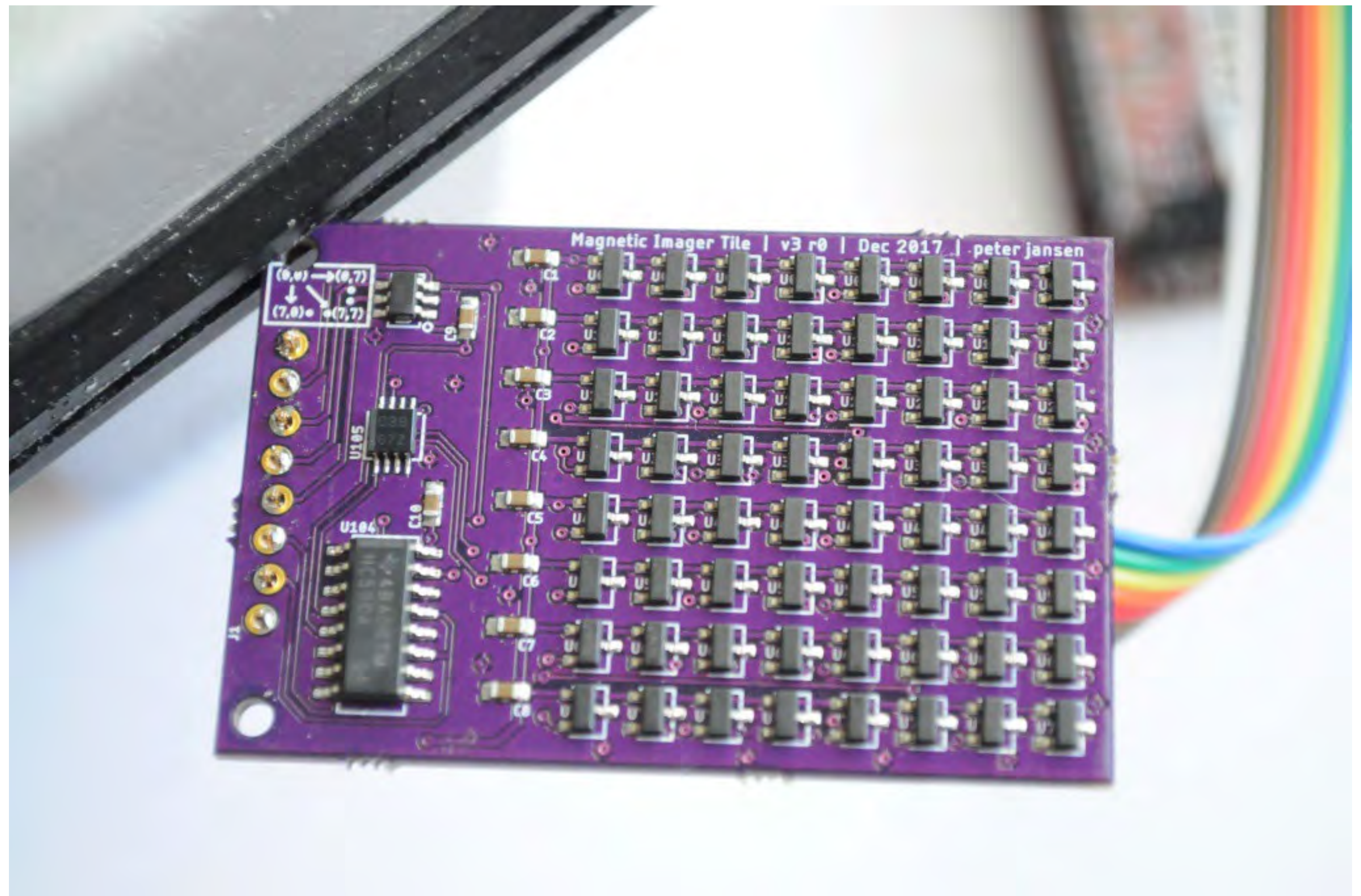


Aims

- We need a device to **monitor the toroidal field** to make sure SOUTH will actually work
- **De-risk** the R&D and manufacturing process

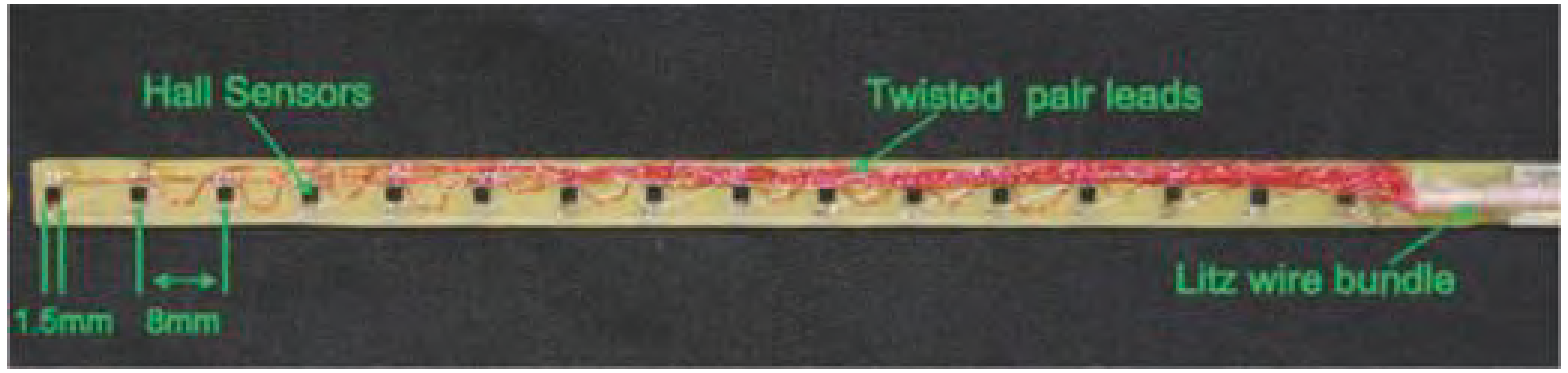


Existing Solutions

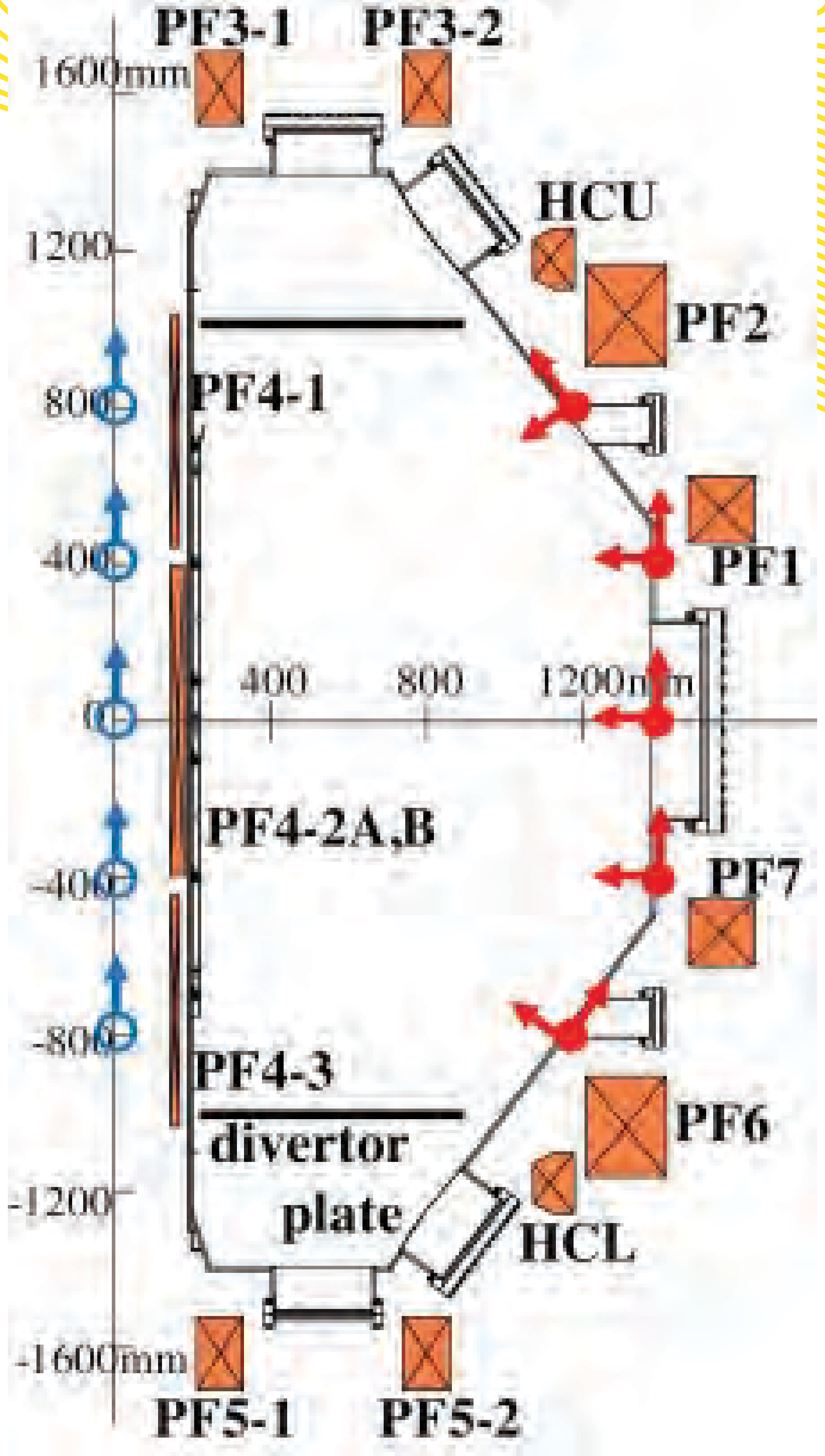


Jansen[1]: educational tool, too slow and no filtering or buffering

Dûran[2]: single, radiation hardened sensor for ITER



Stevenson[3]: Insertable 1D poloidal array with commercial sensors



Hasegawa[4]: QUEST probe locations

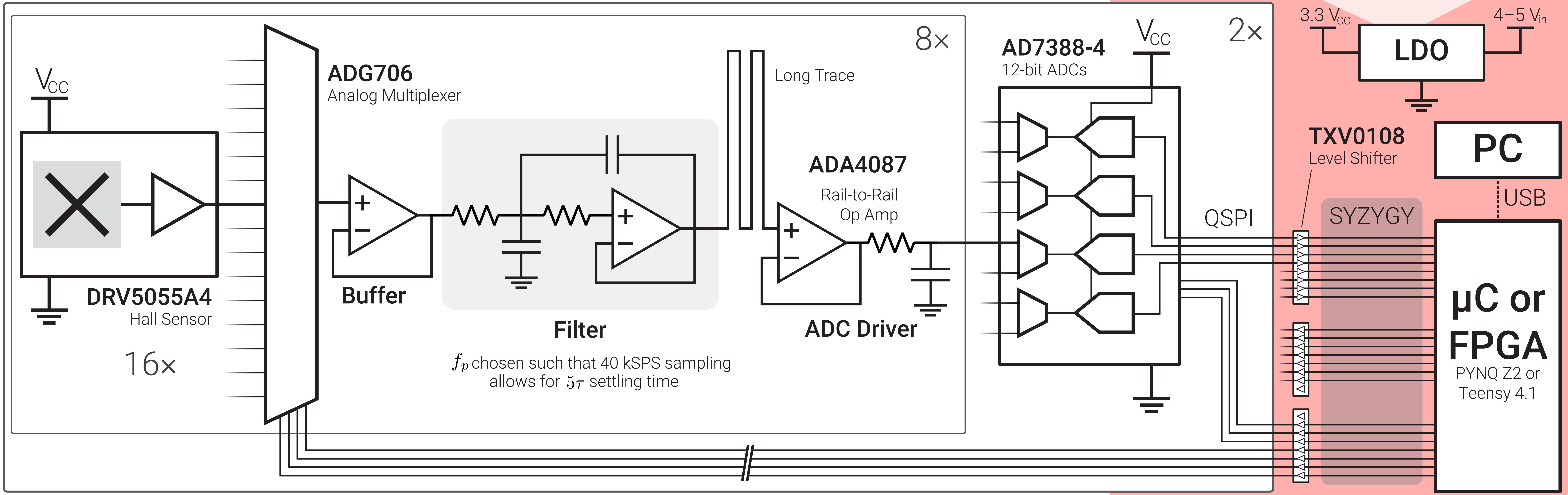
Design Requirements specific **for fusion** and use in **SOUTH**:

- **High Bandwidth** to monitor the dynamic field, >10 kHz is *desirable*
- **High Sensitivity** to catch stray field fields, >1 mT is *desirable*—**EMI resistance**
- **High Dynamic Range** to monitor large fields, >±150 mT *required*
- Must **fit inside SOUTH** through its flange with width of 150 mm

Device Methodology

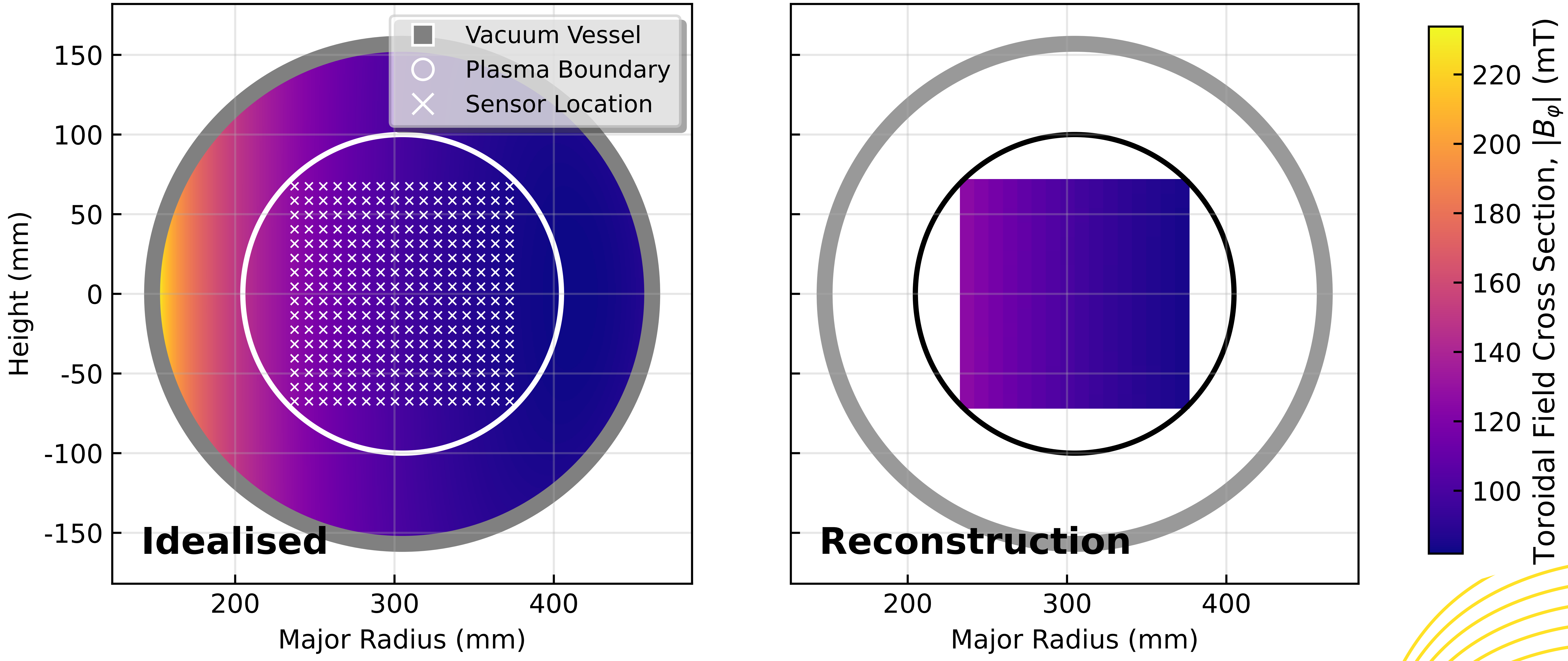
An architecture optimised for **low noise** with **20 kHz** of bandwidth

- 256 sensors in 16x16 grid arranged with 9 mm sensor pitch
- Multiplexed topology: 16:1 into filters, 2:1 into ADC inputs. 2 Discrete ADC ICs with 8 actual silicon ADCs total
- Second-order Sallen-Key filter, plus buffer and ADC driver
- Shared voltage reference across all sensors
- External connection to digital controller (either µC or an FPGA)

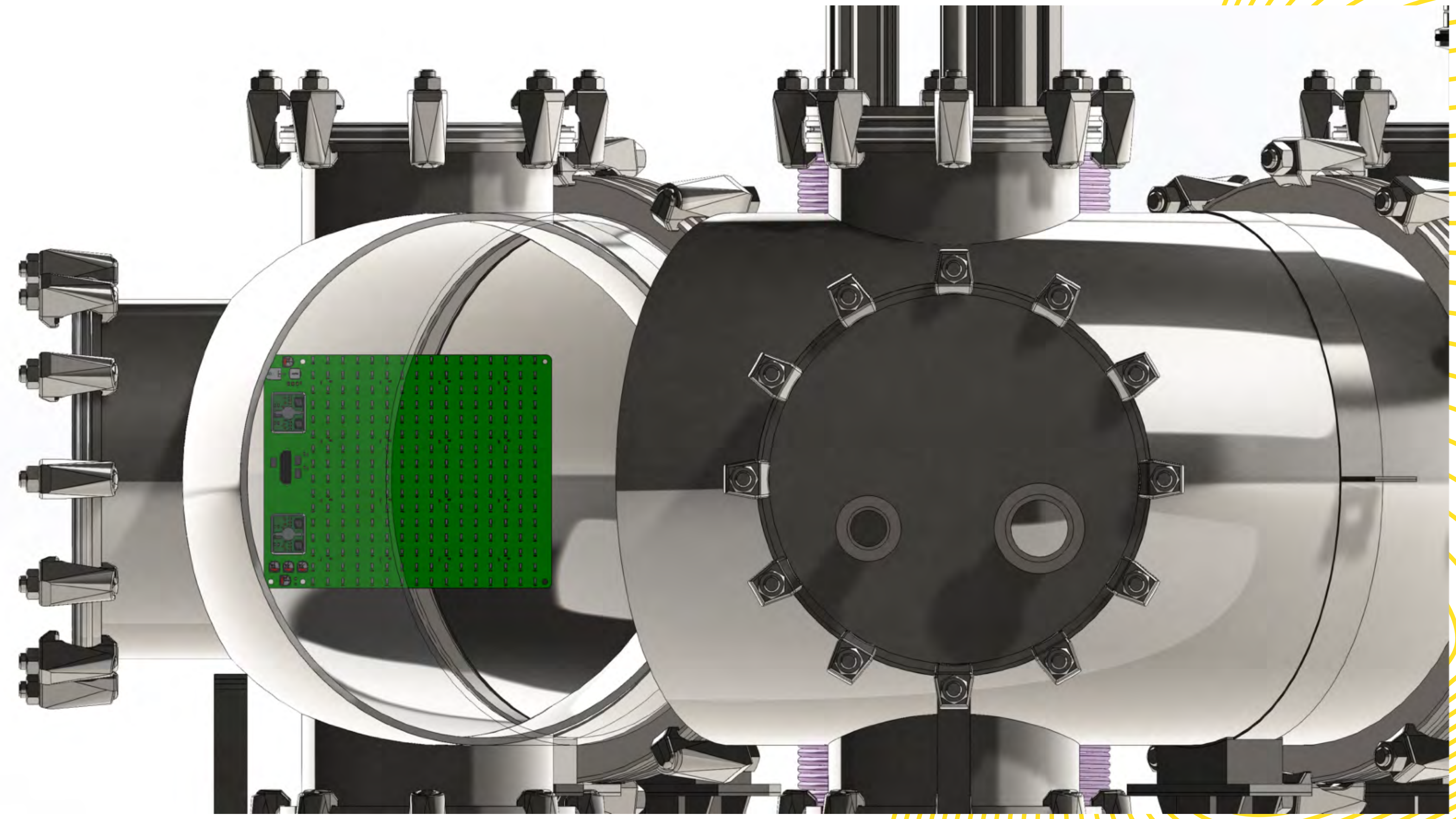


samsh, 2025

Device Methodology



A simulated **toroidal field reconstruction**



samsh, 2025

Results

Specifications

Sensor Layout .... 16×16, 9mm Pitch

Dimensions.......................... 175×142 mm

Dynamic Range.......................... >±150 mT

Bandwidth .... 20 kHz

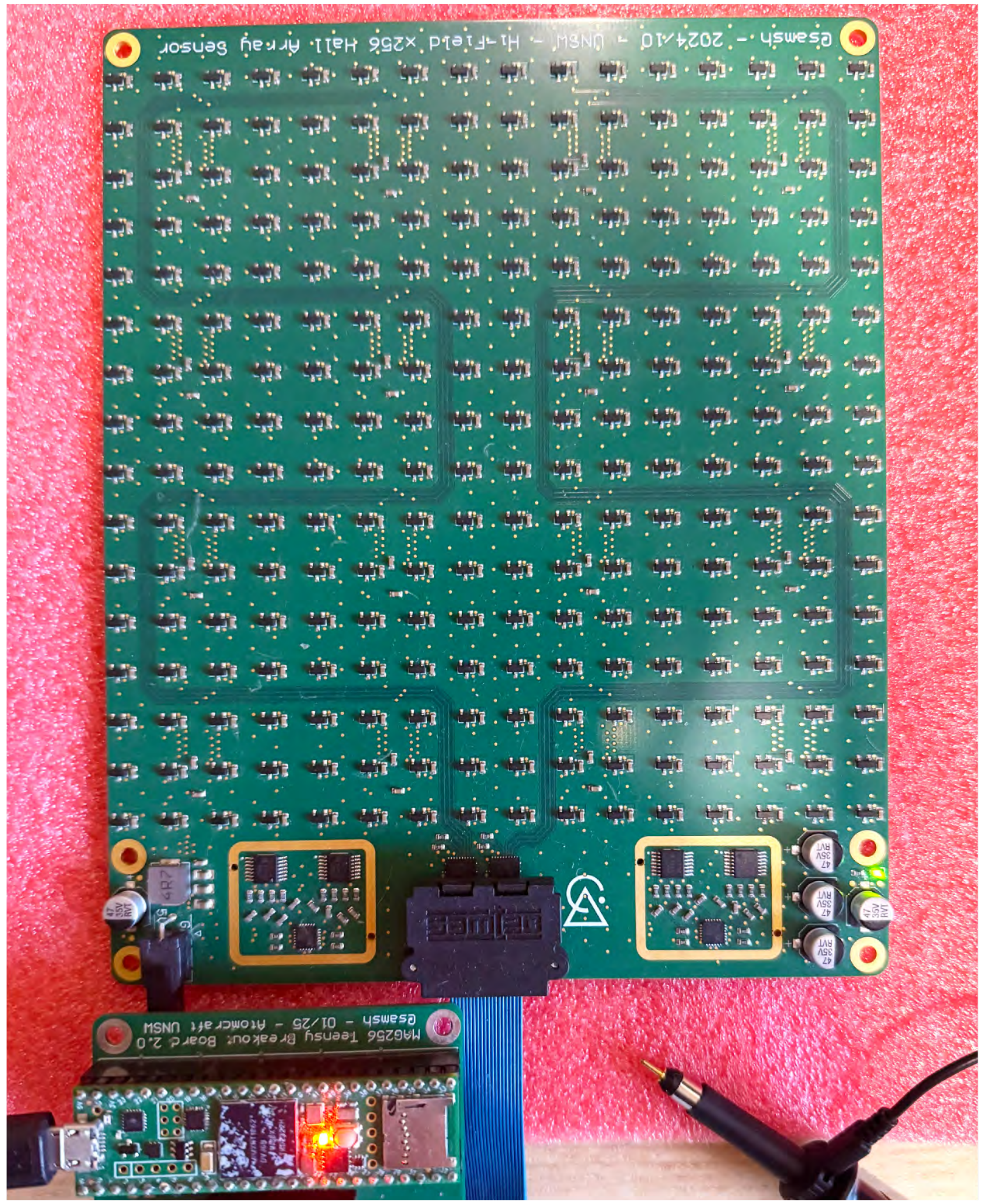
Sampling .... >40 kSPS/sensor

Sensitivity.................................... 0.32 mT

Bits...................................12, ~10.4 ENOB

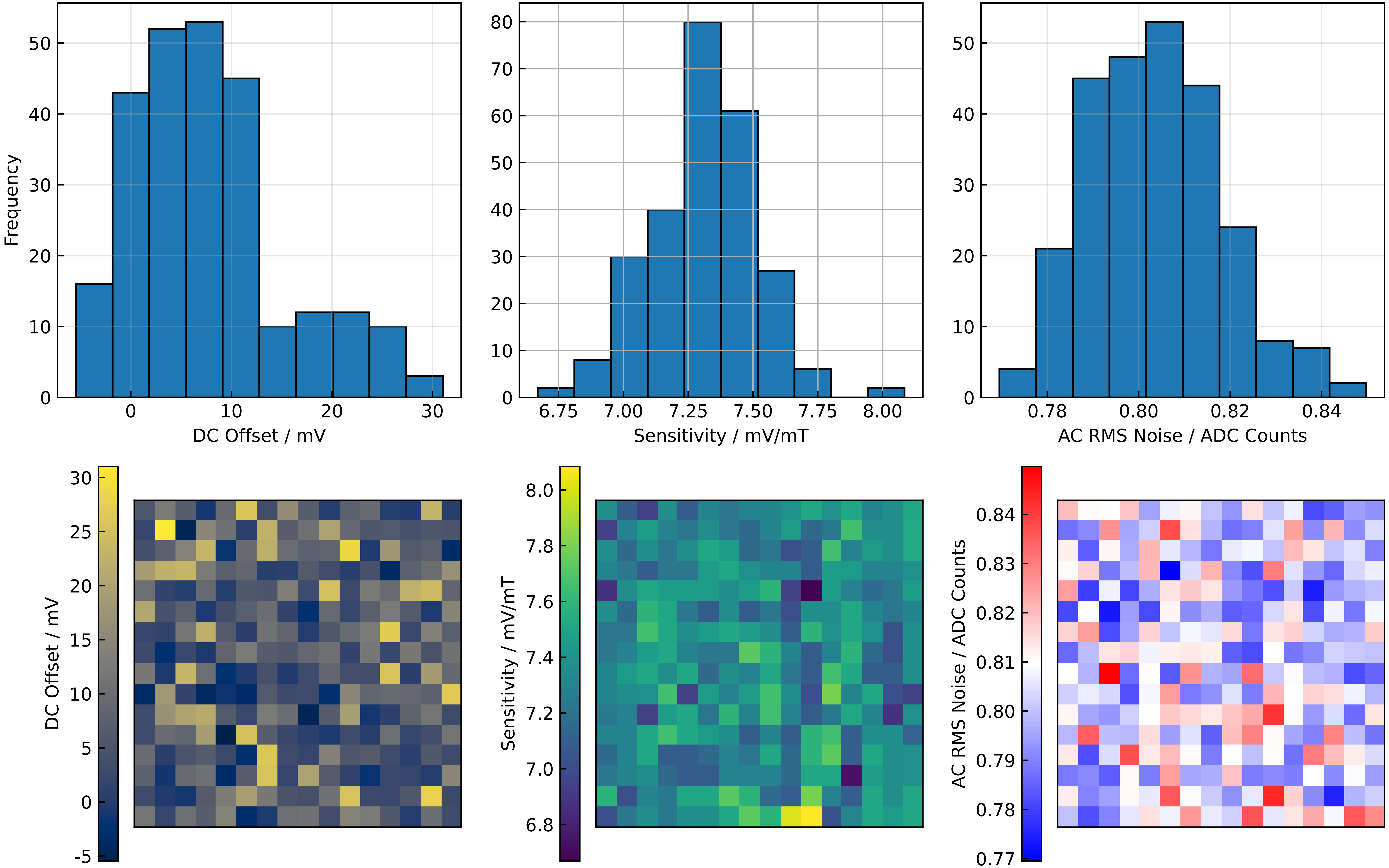
Number of Components .... >600

Power Consumption .... ~7.5 W



Final Device

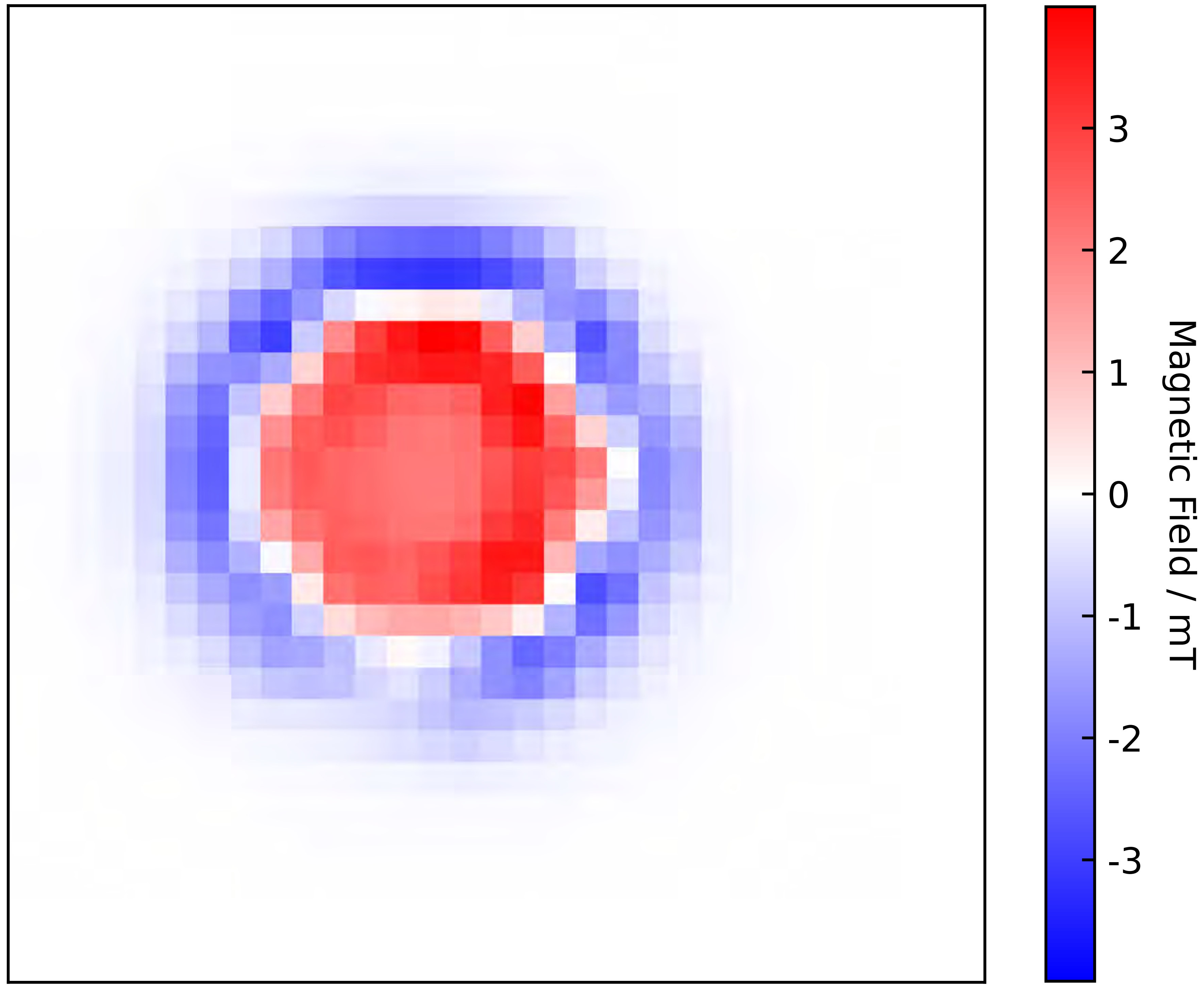
Comprehensive fulfilment of all **Design Goals**



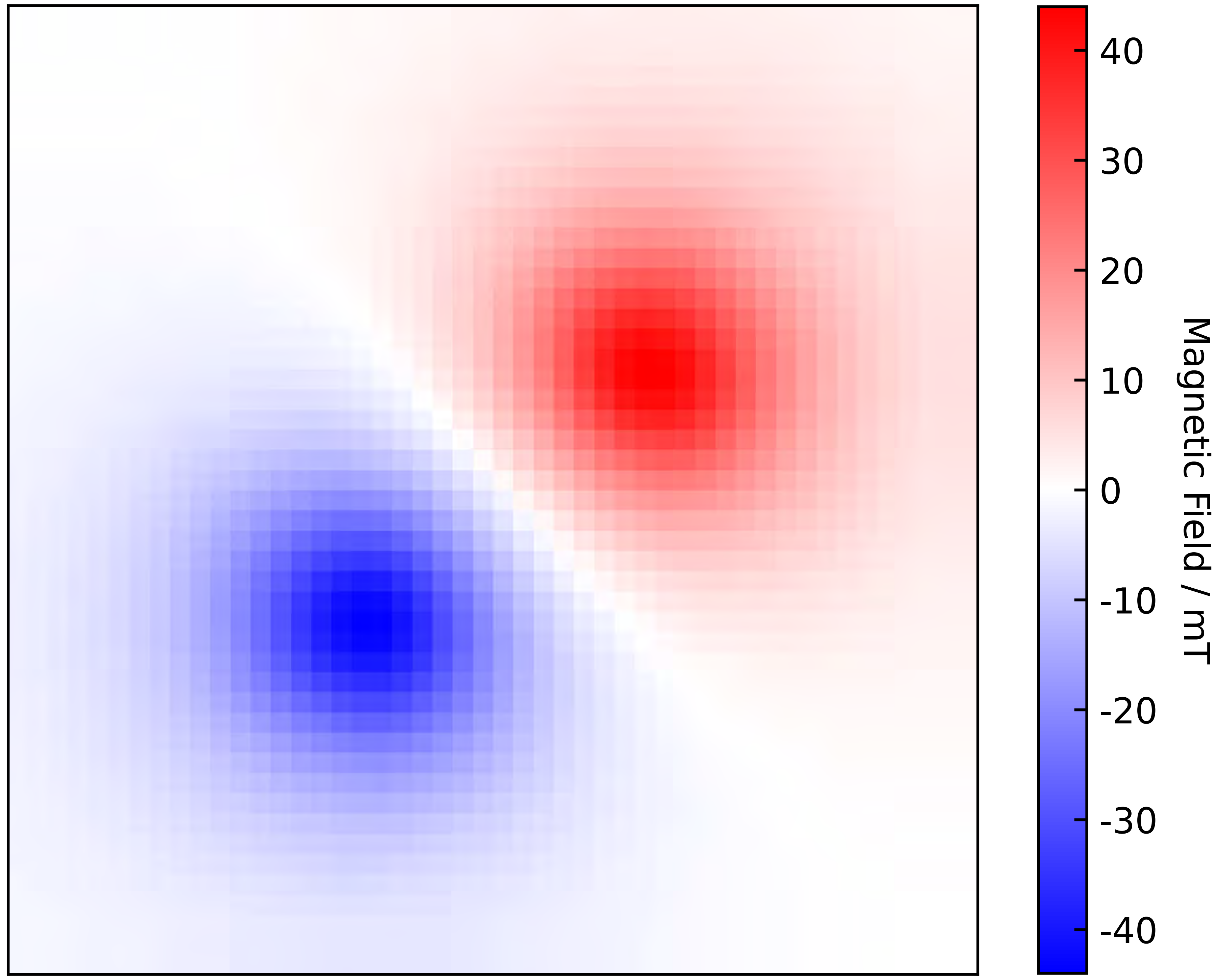
Results

Linear constants and AC RMS Noise

Results



MagSafe Charger



Rare-Earth Bar Magnet

Static Imagery enhanced through **Bicubic Interpolation**

Results

The video here is very similar to
the one at https://youtu.be/sB7mTB_tMe0

Dynamic imagery demonstrated **real-time operation**

What's Next?

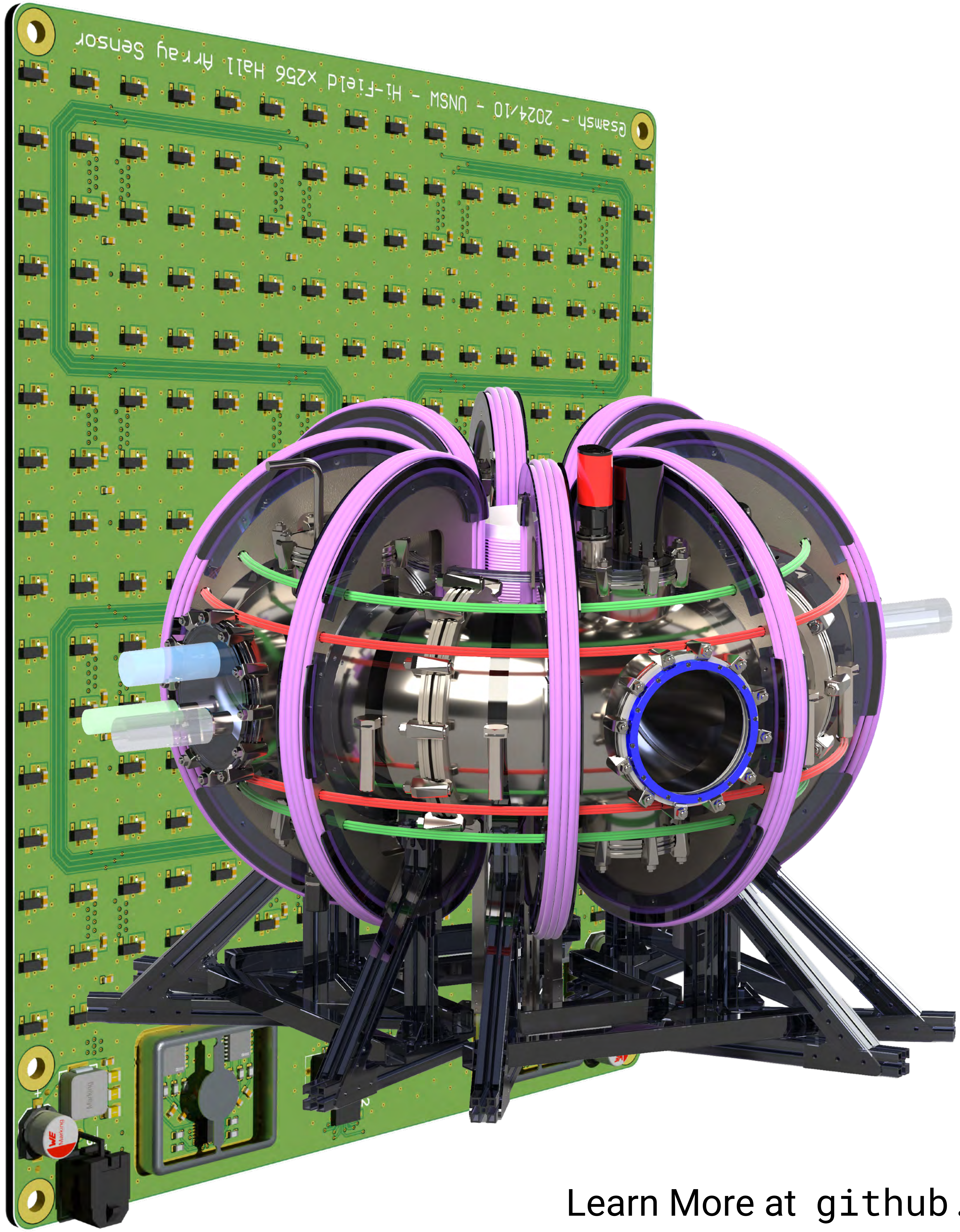
- This work received a **Positive Reception** at the Symposium on Fusion Engineering, in Cambridge, MA; awarded UNSW nuclear innovation centre (UNIC) and Tyree foundation travel grant
- Immediate future priority: development of FPGA frontend to get full 20 kHz bandwidth with >40,000 SPS
- Will be used for SOUTH R&D, 2025–26
- From SOFE: collaboration with SMART
  - Coil alignment for second campaign
  - I will visit them in Seville in October
- Project scope is limited to small devices

The SMART Tokamak[5]
A collaboration between Seville and PPPL

Summary

A **magnetic camera** designed to monitor Tokamak toroidal *B* fields, designed to address the requirements of the upcoming SOUTH Tokamak:

- **20 kHz Bandwidth** achieved through and FPGA and multiple ADCs, or use slower real-time operation with a microcontroller
- **Dynamic range** exceeding of ±150 mT
- **<1 LSB noise** floor facilitates an effective **sensitivity of ≈ 0.32 mT**
- Robust filtering regime to allow for **EMI resistance**
- **Pedagogical** and outreach potential
- Stay tuned for AtomCraft R&D and SMART commissioning!



Thank You!

Learn More at `github.com/Sam-js2/BEThesis` or by scanning the QR code

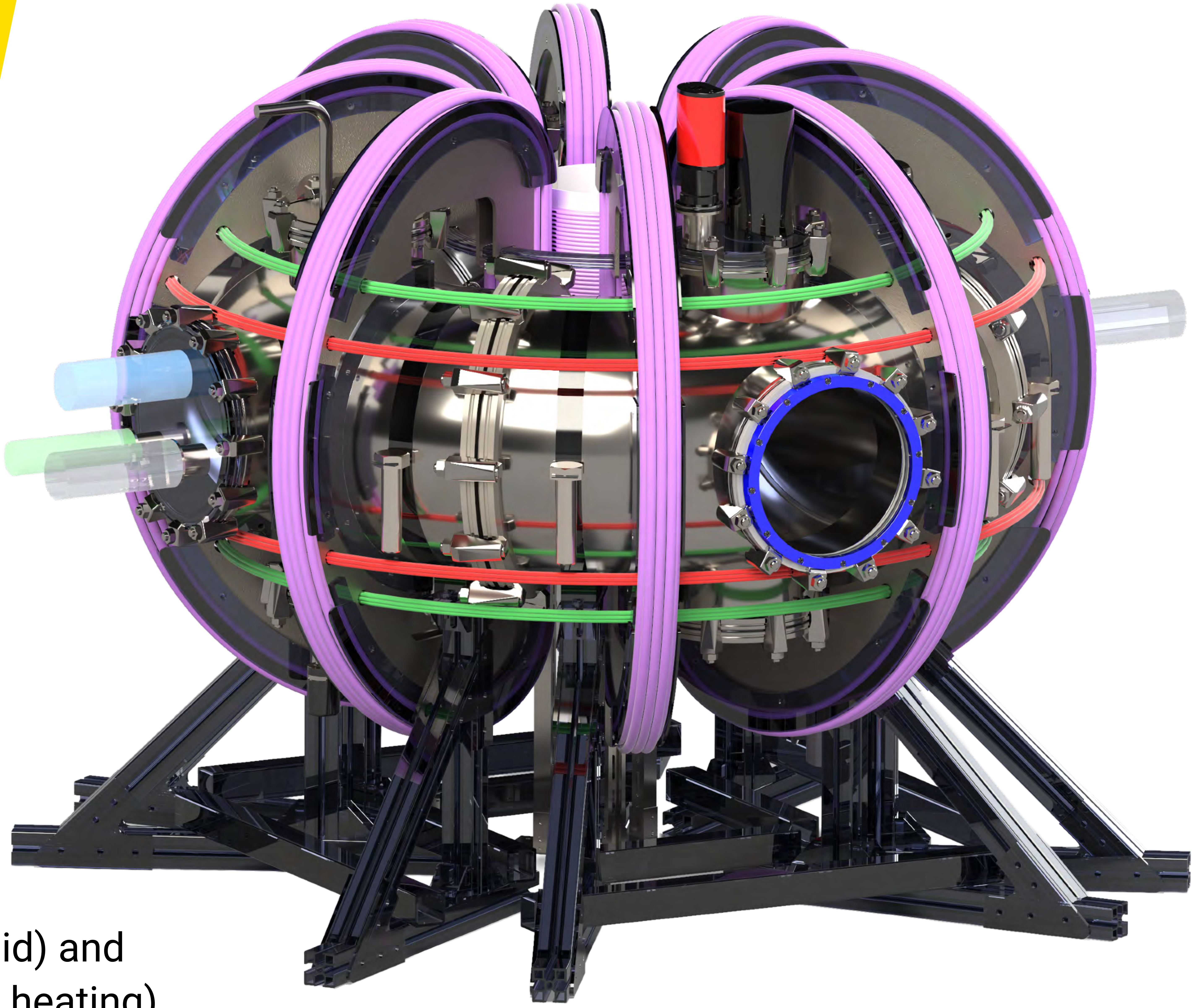
Bibliography

1. Jansen, P., 2018. A Third, High-Speed Magnetic Imager Tile [WWW Document]. Hackerday.io. URL https://hackaday.io/project/18518-iteration-8/log/91551-a-third-high-speed-magnetic-imager-tile (accessed 11.6.24).
2. Ďuran, I., Entler, S., Kohout, M., Kočan, M., Vayakis, G., **2016**. High magnetic field test of bismuth Hall sensors for ITER steady state magnetic diagnostic. *Review of Scientific Instruments* 87, 11D446. https://doi.org/10.1063/1.4964435
3. Stevenson, B.A., Knowlton, S.F., Hartwell, G.J., Hanson, J.D., Maurer, D.A., **2014**. Hall probe measurements of the poloidal magnetic field in Compact Toroidal Hybrid plasmas. *Review of Scientific Instruments* 85, 093502. https://doi.org/10.1063/1.4894209
4. Hasegawa, M., Nakamura, K., Zushi, H., Hanada, K., Fujisawa, A., Tokunaga, K., Idei, H., Nagashima, Y., Kawasaki, S., Nakashima, H., Higashijima, A., **2016**. Current status and prospect of plasma control system for steady-state operation on QUEST. *Fusion Engineering and Design* 112, 699–702. https://doi.org/10.1016/j.fusengdes.2016.04.016
5. Agredano-Torres, M., Garcia-Sanchez, J.L., Mancini, A., Doyle, S.J., Garcia-Munoz, M., Ayllon-Guerola, J., Barragan-Villarejo, M., Viezzer, E., Segado-Fernandez, J., Lopez-Aires, D., Toledo-Garrido, J., Buxton, P.F., Chung, K.J., Garcia-Dominguez, J., Garcia-Franquelo, L., Gryaznevich, M.P., Hidalgo-Salaverri, J., Hwang, Y.S., Leon-Galvan, J.I., Maza-Ortega, J., **2021**. Coils and power supplies design for the SMART tokamak. *Fusion Engineering and Design* 168, 112683. https://doi.org/10.1016/j.fusengdes.2021.112683

Backup: SOUTH

Major Radius	~0.3 m
Minor Radius	30 cm
Peak Temperature	10 eV*
Plasma Current	3 kA
Toroidal on-axis Field	100 mT
Pulse Duration	100 ms

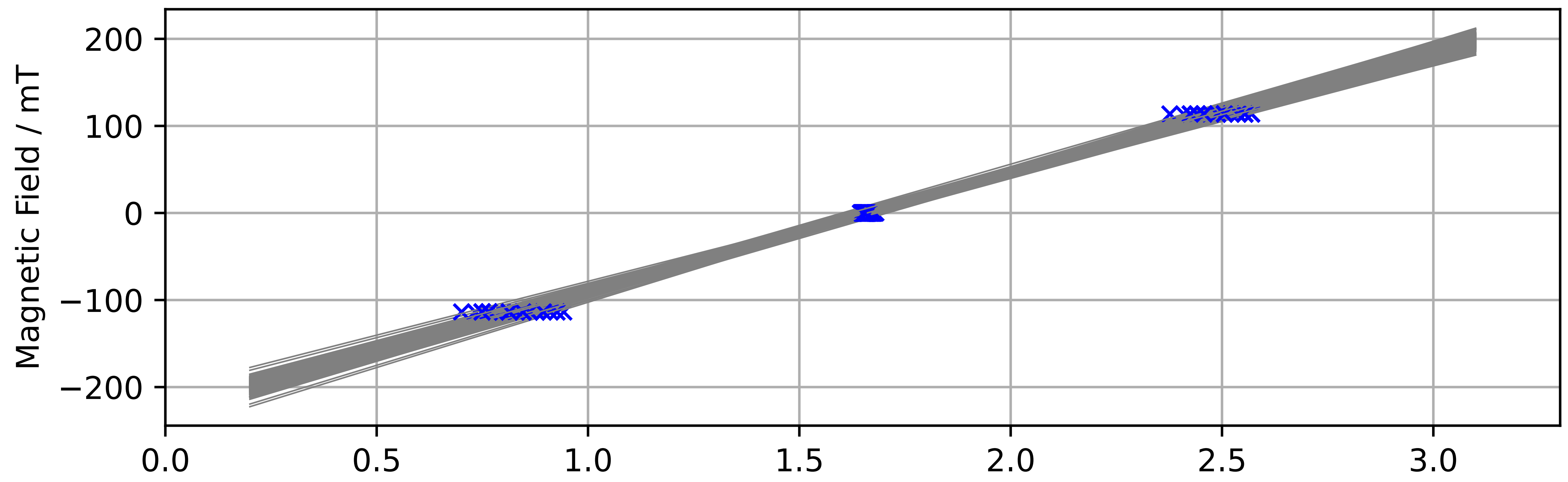
*Thats 120,000 degrees!

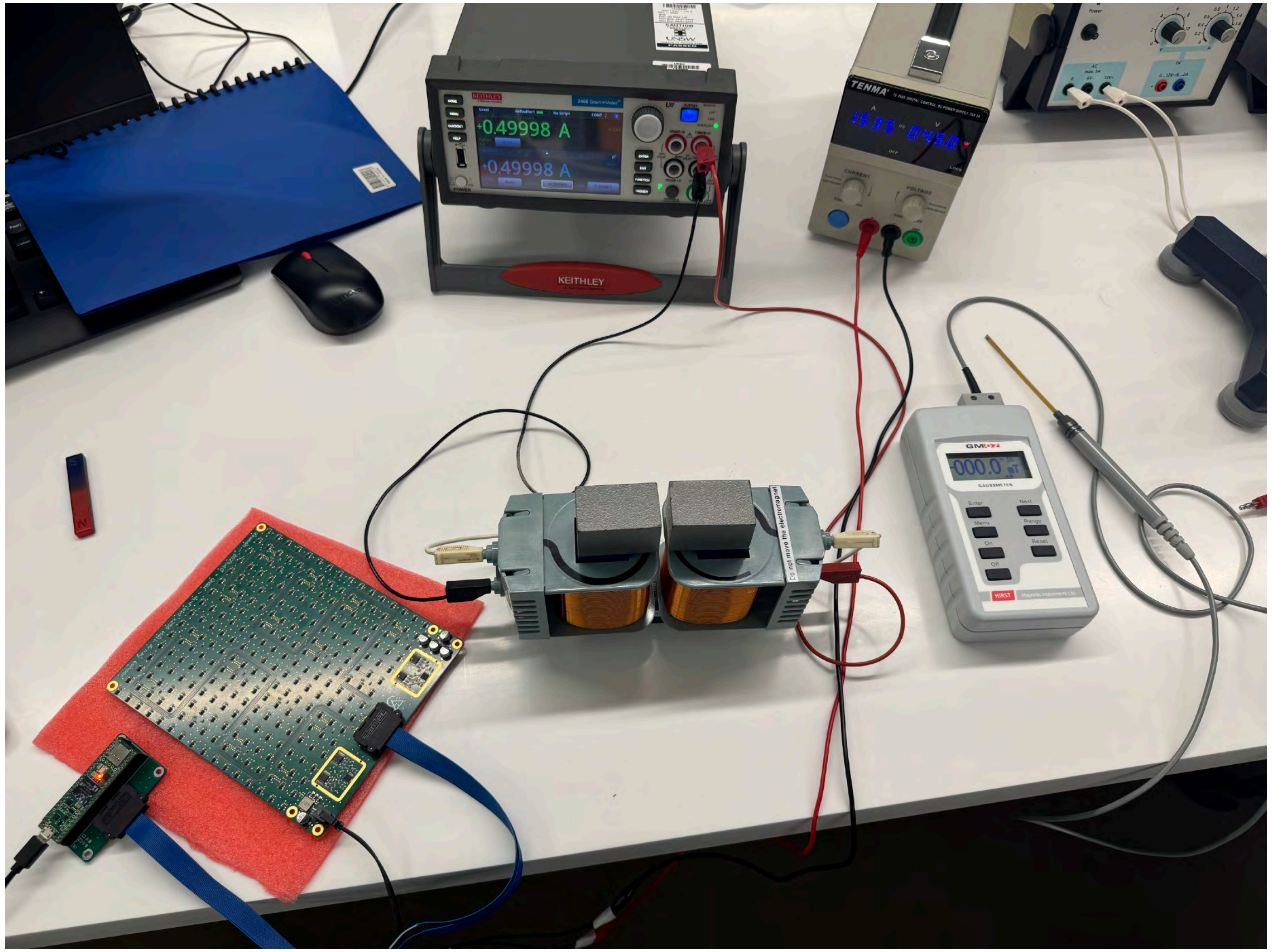


Ionisation with the help of **ohmic heating** (solenoid) and **electron cyclotron resonance heating** (ECRH–RF heating).

Student Operation Undergraduate Tokamak with Hydrogen

Backup: Calibration





Use **ferrite-core electromagnet** powered (extremely precisely) with a **source-meter:**

- ~115 mT field (<1% variations across surface), measured with gauss meter
- Measure aligned and anti aligned (i.e., +/−)
- Gather **lots** of datapoints
- Bin into histogram, take highest frequency bin
- Linear regression on +115/0/−115 mT
- Linearity is already guaranteed by manufacturer, just getting sensitivity